

2020 概率论与数理统计练习卷 III 解答

本试题可能用到的常数:

$$u_{0.95}=1.645; u_{0.975}=1.960; u_{0.900}=1.280; \chi_{0.1}^2(19)=27.204$$

一、填空题 (每小题 3 分, 共 30 分)

1. 0.25

2. 24/5

3.
$$f_{Y|X}(y|x) = \frac{f(x,y)}{f_X(x)} = \begin{cases} \frac{1}{2x}, & -x < y < x, \\ 0, & \text{其它 } y. \end{cases}$$

4. 0.1

5. 23.4

6. 0.95

7. $\pm\sqrt{2}$, $t(2)$

8. $\hat{\mu}_2$ (注: 其中 $(D(\hat{\mu}_1) = \frac{5}{9}D(X), \hat{\mu}_2 = \frac{1}{2}D(X))$)

9. $(\bar{X} \pm \frac{\sigma}{\sqrt{n}} z_{\alpha/2}) = (580 \pm 9.8) = (570.2, 589.8)$

二、(10 分) 解: 令血糖值为 X , 事件 A 为患糖尿病, 事件 B 为一次检查中血糖值超过 7, (2 分)

则 $P(A) = P\{X > 7\} = P\{\frac{X-5}{1.56} > \frac{7-5}{1.56}\} = 1 - \Phi(1.28) = 0.1$, (5 分) 由贝叶斯公式有

$$\begin{aligned} P(A|B) &= \frac{P(B|A)P(A)}{P(B|A)P(A) + P(B|\bar{A})P(\bar{A})} \\ &= \frac{0.95 \times 0.1}{0.95 \times 0.1 + 0.06 \times 0.9} = \frac{0.095}{0.149} = 0.64 \end{aligned} \quad (10 \text{ 分})$$

三、解 函数 $y = \sin x$ 在区间 $(0, \pi/2)$ 上严格单调, 其反函数和反函数导数分别为

$$h(y) = \arcsin y \quad (0 < y < 1), \quad h'(y) = \frac{1}{\sqrt{1-y^2}}$$

$$\begin{aligned} \therefore f_Y(y) &= |h'(y)| \cdot f_X(h(y)) \\ &= \begin{cases} \left| \frac{1}{\sqrt{1-y^2}} \right| \cdot \frac{2}{\pi}, & 0 < y < 1, \\ 0, & \text{other} \end{cases} = \begin{cases} \frac{2}{\pi\sqrt{1-y^2}}, & 0 < y < 1, \\ 0, & \text{other} \end{cases} \end{aligned}$$

四、(10 分) 解: $f_Z(z) = \int_{-\infty}^{+\infty} f(x, z-x) dx = \int_0^1 f(x, z-x) dx, \dots\dots 3 \text{ 分}$

当 $z \leq 0, \text{ or } z \geq 2$ 时, $f(x, z-x) = 0. \Rightarrow f_Z(z) = 0$

当 $0 < z \leq 1$ 时,

$$f_Z(z) = \int_{z/2}^z 3x dx = \frac{9}{8} \cdot z^2; \dots\dots 6 \text{ 分}$$

当 $1 < z < 2$ 时, $f_Z(z) = \int_{z/2}^1 3x dx = \frac{3}{2}(1 - \frac{z^2}{4}) \dots\dots 8 \text{ 分}$

$$\therefore f_Z(z) = \begin{cases} \frac{9}{8} \cdot z^2 & , 0 < z \leq 1 \\ \frac{3}{2} (1 - \frac{z^2}{4}) & , 1 < z < 2 \\ 0 & , \text{其它} \end{cases} \dots\dots 10 \text{ 分}$$

五、(10 分) 解 $E(X) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xf(x, y) dx dy = 12 \int_0^1 dx \int_0^x xy^2 dy = 4 \int_0^1 x^4 dx = \frac{4}{5}$

$$E(Y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} yf(x, y) dx dy = 12 \int_0^1 dx \int_0^x y^3 dy = 3 \int_0^1 x^4 dx = \frac{3}{5}$$

又

$$E(XY) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xyf(x, y) dx dy = 12 \int_0^1 dx \int_0^x xy^3 dy = 3 \int_0^1 x^5 dx = \frac{1}{2}$$

所以

$$\text{Cov}(X, Y) = E(XY) - E(X) \cdot E(Y) = \frac{1}{2} - \frac{4}{5} \cdot \frac{3}{5} = \frac{1}{50}$$

六、(15 分) 解: 1) 似然函数为: $L(\theta) = \prod_{i=1}^n f(x_i) = \prod_{i=1}^n \frac{2x_i}{\theta} e^{-x_i^2/\theta} = \frac{\prod_{i=1}^n 2x_i}{\theta^n} e^{-\frac{1}{\theta} \sum_{i=1}^n x_i^2} \dots\dots 3 \text{ 分}$

对数似然函数为: $l(\theta) = \ln L(\theta) = \sum_{i=1}^n \ln 2x_i - n \ln \theta - \frac{1}{\theta} \sum_{i=1}^n x_i^2 \dots\dots 5 \text{ 分}$

$$\frac{d}{d\theta} l(\theta) = -\frac{n}{\theta} + \frac{1}{\theta^2} \sum_{i=1}^n x_i^2 = 0 \Rightarrow \hat{\theta} = \frac{1}{n} \sum_{i=1}^n x_i^2 \dots\dots 10 \text{ 分}$$

2) $E(X^2) = \int_{-\infty}^{\infty} x^2 f(x) dx = \int_0^{\infty} x^2 \frac{2x}{\theta} e^{-x^2/\theta} dx = \int_0^{\infty} x^2 \frac{1}{\theta} e^{-x^2/\theta} dx = \int_0^{\infty} t \frac{1}{\theta} e^{-t/\theta} dt = \theta$ 所以是无偏估计。 15 分

七、(15 分) 解 依题意, 要求检验 $H_0: \sigma^2 \leq \sigma_0^2 = 450$, $H_1: \sigma^2 > \sigma_0^2$...4 分

采用单边 χ^2 检验法, 求得检验统计量为 $\frac{(n-1)S^2}{\sigma_0^2}$ 8 分

H_0 的拒绝域为 $\frac{(n-1)S^2}{\sigma_0^2} > \chi_{\alpha}^2(n-1) = \chi_{0.1}^2(19) = 27.204$...10 分

由数据算得 $\frac{(n-1)S^2}{\sigma_0^2} = \frac{19 \times 663.2}{450} = 28.002 > 27.204$ 故拒绝 H_0 , 即认为这批产品的方差不符合要求。...15 分